Buddhi Kothalawala

PhD Candidate & Senior Software Engineer

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Summary

Ph.D. candidate in Computer Science (ANU) specialising in branch-and-bound algorithms, combinatorial optimisation, and constraint programming, with peer-reviewed publications at SIGMOD 2026, CP 2025, and ADC 2025. Research background spans combinatorial optimisation and machine learning, including deep learning work on low-resource NLP using RNNs, bidirectional LSTMs, and online learning algorithms recognised with the ICTA Award for Best Computer Science Final Year Research Project. Brings 3+ years of production software engineering experience at WSO2, bridging deep algorithmic research with real-world system development.

Education

The Australian National University, Canberra, Australia
Ph.D. in Computer Science (Thesis under review) (Expected 2026)

- **Thesis**: Algorithmic Advances in Exact Maximum Common Subgraph (MCS) Computation
- **Research Focus**: Novel branch-and-bound algorithms enhanced with reinforcement learning
- **Funding**: ANU PhD Scholarship (International) + HDR Fee Remission Merit Scholarship

University of Colombo School of Computing, Colombo, Sri Lanka
Bachelor of Science in Computer Science (Honours), 2019

- **Class**: First Class Honours
- **Thesis**: Online Learning for Solving Data Availability Problem in Natural Language Processing
- **Recognition**: ICTA Award for Best Computer Science Final Year Research Project (2018)

IELTS Academic, Overall score 7.0, 2021

Research & Publications

Kothalawala, B., Koehler, H., & Farhan M. Accelerating Maximum Common Subgraph Computation by Exploiting Symmetries. (ACM Special Interest Group on Management of Data (SIGMOD) 2026 (CORE A^*, Accepted) [Paper, C++ Code])

- Developed a symmetry breaking framework for Maximum Common Subgraph algorithms that substantially outperforms state-of-the-art methods.
- Demonstrated significant speedups on standard MCS benchmarks through comprehensive symmetry-aware pruning, solving more instances with reduced computation time and search space.

Kothalawala, B., Koehler, H., Wang Q. & Farhan M. (2025). From Exploratory Heuristics to Exact Search: Accelerating Maximum Common Subgraph Algorithms. (Australasian Database Conference (ADC)). [Paper, C++ Code].

- Proposed a novel heuristic mechanism to escape local optima in depth-first branch-and-bound algorithms for maximum common subgraph discovery.
- Developed a principled transition criterion from heuristic to exact search, with a dynamic mechanism balancing exploration and exhaustive search.

Kothalawala, B., Koehler, H., & Wang Q. (2025). Learning to Bound for Maximum Common Subgraph Algorithms. (The 31st International Conference on Principles and Practice of Constraint Programming (CP)). (CORE A) [Paper, C++ Code].

- Developed a novel bound for NP-hard optimisation problems activated by a reinforcement learning framework, demonstrating greater empirical performance than existing branch-and-bound approaches.

Kothalawala, B., Weerasinghe, R., & Kumarasinghe, P. (2019). Online Learning for Solving Data Availability Problem in Natural Language Processing. NL4AI@AI*IA. (Paper) (Python Code)

- Addressed model retraining challenges for the low-resource language Sinhalese
- Applied online learning algorithms alongside conditional random fields (CRFs), RNNs, and bi-directional LSTMs to overcome data scarcity in NLP applications

Kothalawala, B., Koehler, H., & Farhan M. Maximum Common Subgraph Counting with Symmetry Breaking. *Under Review.*

- Developed a counting framework that corrects counts pruned by symmetry breaking and enumerates pruned subgraphs, enabling accurate enumeration under symmetry-aware search.

Research Interests Combinatorial Optimisation, Branch-and-Bound Algorithms, Reinforcement Learning, Constraint Programming, Online Machine Learning, Computational Complexity

Experience

UNSW Canberra at the Australian Defence Force Academy

Research Assistant (Part Time) **June 2026 – October 2026**

- Conducted research on graph theory and subgraph-based algorithms applied to supply chain networks.
- Performed theoretical analysis and experimental evaluation of these algorithms in C++ and Python.

The Australian National University, Canberra, Australia

Academic Tutor **February 2023 – Present**

- Taught and mentored over 150 students in Statistical Machine Learning (COMP4670/8600), Relational Databases (COMP2400/6240), TechLauncher (COMP3500/COMP4500/COMP8715), and Data Wrangling (COMP3430/8430).
- Helped with a final-year software project in the Tech Launcher course.

Ph.D. Candidate & Researcher **June 2022 – June 2026**

- Designed, coded, tested, and deployed high-performance software systems in C++ for the Maximum Common Subgraph (MCS) problem, applying advanced data structures and algorithm design to achieve state-of-the-art benchmark results.
- Researched and designed a new symmetry-breaking software framework; wrote clean, scalable code with correctness guarantees; accepted at SIGMOD 2026 (CORE A*).
- Designed a reinforcement learning module that activates a novel upper bound within a branch-and-bound framework, identifying and exploiting modification opportunities in an existing solver to improve runtime performance; published at CP 2025 (CORE A).
- Designed and implemented a new heuristic software component with a dynamic transition mechanism, extending existing branch-and-bound programs; published at ADC 2025.
- Troubleshot, debugged, and verified software correctness across multiple program versions; maintained code quality through systematic testing protocols and version control using Git and GitHub.
- Conducted extensive code profiling to identify and resolve performance bottlenecks.

WSO2, Colombo, Sri Lanka

Senior Software Engineer **November 2020 – June 2022**

- Designed, coded, and deployed gRPC, I/O, and Time standard libraries for the Ballerina programming language, writing clean, scalable code and maintaining software security across multiple release cycles.
- Analysed existing gRPC programs for modification and implemented improvements: deadline feature, call security, compiler plugin, backward compatibility fixes, annotation and Any-type support, and xDS API (proof-of-concept).
- Researched and designed an I/O API redesign with streaming capabilities; implemented new JSON and XML APIs according to revised software design specifications.
- Implemented UTC and time zone handling APIs, verified and deployed Time library.
- Led standard library migration project: modified and migrated Ballerina by Examples.
- Created technical documentation and trained developer community through the first Ballerina Language Community Call (1,200+ views).
- Authored technical articles on Medium and DZone (20,000+ combined views); authored DZone article on gRPC compression support in Go, Java, and Ballerina.
- Received Sustained Outstanding Contribution Award (2020).

Software Engineer **February 2019 – October 2020**

- Designed, coded, tested, and deployed the Siddhi Kubernetes operator in Go, troubleshot and debugged deployment issues across Docker and Kubernetes environments.
- Analysed existing Siddhi programs and modified them to integrate NATS and NATS Streaming for event-driven messaging.
- Implemented the Siddhi Docker Kubernetes export feature, writing and verifying scalable code for cloud-native deployment.
- Created technical documentation, including interactive user examples and Siddhi deployment guides.
- Led the migration of Ballerina from a monolithic to a multi-repository software architecture; established CI/CD deployment pipelines that reduced integration overhead and release cycle times.
- Received Sustained Outstanding Contribution Award (2019).

Software Engineer Intern **August 2017 – January 2018**

- Designed, coded, tested, and deployed a full-stack internal web application for license and repository management using React (JavaScript/HTML), Ballerina, and Java across frontend, backend, and integration layers.
- Liaised with senior developers on project scope and troubleshot bugs.

Technical Skills

- **Languages:** C++, Go, Java, Python, Ballerina, SQL
- **Machine Learning & AI:** PyTorch, Reinforcement Learning, Online Learning, Statistical Machine Learning, Constraint Programming, RNN, LSTM
- **Databases:** PostgreSQL, MySQL
- **Cloud & DevOps:** Kubernetes, Docker, K8s Operators, gRPC, Protobuf, NATS, Linux, Performance Tuning
- **Development Tools:** Git, GitHub, Gradle, Maven, CI/CD, Code Reviews, Technical Documentation

- **Specialisations:** Branch-and-Bound Algorithms, Combinatorial Optimisation, Siddhi Stream Processing, Software Architecture, API Integration

Awards and Recognition

- **ANU PhD Scholarship (International)** - The Australian National University
- **HDR Fee Remission Merit Scholarship** - The Australian National University
- **Sustained Outstanding Contribution Awards** - WSO2 (2019, 2020)
- **ICTA Award** - Best Computer Science Final Year Research Project (2018)

Professional Service & Community Engagement

- **Open Source**: Core contributor to Ballerina programming language and Siddhi streaming ecosystem; StackOverflow community contributor (1K+ reputation)
- **Community Leadership**: Conducted the first Ballerina Language Community Call (1.2K+ views); published technical articles on Medium and DZone with 20K+ views
- **Leadership**: President, WSO2 Toastmasters Club (2020/21); completed three levels of the Toastmasters Dynamic Leadership Pathway
- **Research Supervision**: Mentored a final year project on Compressing Bi-LSTM-based Contextualized Word Embeddings; guided junior developers at WSO2 on best practices and career development
- **Academic Tutoring**: Tutored 150+ students across 4 advanced courses at ANU
- **Conference Presentations**: Presented research at CP 2025 and ADC 2025
- **Volunteer**: Solid Hackathon 2026, Australian National University

References

References available upon request.